

PYTHON LOGICAL QUESTIONS — CONDITIONAL STATEMENTS

Q1

x = 5

if x > 3:

print("A")

else:

print("B")

Q2

x = 3

if x:

print("Yes")

else:

print("No")

Q3

x = 0

if x:

print("True")

else:

print("False")

Q4

x = -5

if x > 0:

print("Positive")

else:

print("Not Positive")

Q5

x = 10

if x < 5:

print("Low")

elif x < 15:

print("Medium")

else:

print("High")

Q6

x = 5

if x > 10:

print("A")

elif x > 3:

print("B")

elif x > 1:

print("C")

else:

print("D")

Q7

x = 7

if x > 10:

print("X")

if x > 5:

print("Y")

else:

```
print("Z")
```

Q8

```
x = 5
```

```
y = 10
```

```
if x > 3 and y < 15:
```

```
    print("True")
```

```
else:
```

```
    print("False")
```

Q9

```
x = 0
```

```
if x == 0 or x > 5:
```

```
    print("Yes")
```

```
else:
```

```
    print("No")
```

Q10

```
x = 5
```

```
if not x:
```

```
    print("Zero")
```

```
else:
```

```
    print("Non-Zero")
```

Q11

```
x = "False"
```

```
if x:
```

```
    print("True")
```

else:

print("False")

Q12

x = None

if x:

print("Yes")

else:

print("No")

Q13

x = []

if x:

print("List")

else:

print("Empty")

Q14

x = 5

if x == 5 == True:

print("Yes")

else:

print("No")

Q15

x = 10

if x > 5:

if x < 15:

```
    print("A")
else:
    print("B")
else:
    print("C")
```

Q16

```
x = 3
if x > 5:
    print("A")
else:
    if x > 2:
        print("B")
    else:
        print("C")
```

Q17

```
x = 10
if x > 5:
    print("A")
    if x > 15:
        print("B")
else:
    print("C")
```

Q18

```
x = 5
if x > 3:
```

```
    print("A")
    print("B")
else:
    print("C")
```

Q19

```
x = 5
if x > 3:
    print("A")
if x > 4:
    print("B")
else:
    print("C")
```

Q20

```
x = 2
if x > 5:
    print("A")
elif x > 3:
    print("B")
elif x > 1:
    print("C")
else:
    print("D")
```

Q21

```
x = 5
if x > 2:
```

```
print("A")  
if x > 10:  
    print("B")  
else:  
    print("C")  
else:  
    print("D")
```

Q22

```
x = 0  
if x == False:  
    print("Zero")  
else:  
    print("Non-Zero")
```

Q23

```
x = True  
if x == 1:  
    print("Yes")  
else:  
    print("No")
```

Q24

```
x = ""  
if not x:  
    print("Empty")  
else:  
    print("Not Empty")
```

Q25

x = 5

if x:

print("A")

elif x == 5:

print("B")

else:

print("C")

PRACTICE QUESTIONS — CONDITIONAL STATEMENTS (PYTHON)

Q1

Write a program to check whether a number is positive or negative.

Q2

Write a program to check whether a number is even or odd.

Q3

Write a program to check whether a person is eligible to vote (age $\geq$ 18).

Q4

Write a program to check whether a number is zero or non-zero.

Q5

Write a program to check whether a number is greater than 100.

Q6

Write a program to check whether a number is:

- Positive
 - Negative
 - Zero
-

Q7

Write a program to find the largest of two numbers.

Q8

Write a program to find the largest of three numbers.

Q9

Write a program to print:

- "Pass" if marks ≥ 40
 - "Fail" otherwise
-

Q10

Write a program to assign grades:

- A $\rightarrow$ marks ≥ 90
 - B $\rightarrow$ marks ≥ 75
 - C $\rightarrow$ marks ≥ 60
 - D $\rightarrow$ marks ≥ 40
 - F $\rightarrow$ otherwise
-

Q11

Write a program to check whether a number is divisible by both 3 and 5.

Q12

Write a program to check whether a number is divisible by 3 or 7.

Q13

Write a program to check whether a year is a leap year.

Q14

Write a program to check whether a character is a vowel or consonant.

Q15

Write a program to check whether a person is:

- **Child (age < 13)**
 - **Teenager (13–19)**
 - **Adult (20+)**
-

Q16

What will be the output?

x = 0

if x:

print("Yes")

else:

print("No")

Q17

Predict the output:

x = "False"

if x:

print("True")

else:

print("False")

Q18

What is the output?

x = []

if x:

print("List")

else:

print("Empty")

Q19

Predict the output:

x = 5

if x > 3:

print("A")

if x > 4:

print("B")

else:

print("C")

Q20

What will be printed?

x = 10

if x > 5:

if x < 8:

print("A")

else:

print("B")

Q21

Write a program to check whether a number is a multiple of 4 but not a multiple of 8.

Q22

Write a program to check whether a triangle is valid using 3 sides.

Q23

Write a program to calculate electricity bill:

- First 100 units → free
 - Next 100 units → ₹5/unit
 - Above 200 units → ₹10/unit
-

Q24

Write a program to check whether a character is:

- Uppercase letter
 - Lowercase letter
 - Digit
 - Special character
-

Q25

Write a program to simulate a simple login system:

- Correct username & password → "Login Successful"
 - Otherwise → "Access Denied"
-

Q 1 Number Classification (Multiple Conditions)

Write a program that classifies a number as:

- Positive Even
- Positive Odd

- Negative Even
 - Negative Odd
 - Zero
-

Q 2 Smart Calculator (No Loops)

Input two numbers and an operator (+, -, *, /).
Perform the operation using only if-elif-else.

Q 3 Exam Result Analyzer

Input marks of 3 subjects:

- If any subject < 40 → Fail
 - Else if average ≥ 75 → Distinction
 - Else → Pass
-

Q 4 Leap Year with Nested Logic

Check leap year using:

- Divisible by 400 → Leap
- Divisible by 100 → Not Leap
- Divisible by 4 → Leap
- Else → Not Leap

(Use nested if)

Q 5 Triangle Type Detector

Given 3 sides, check:

1. Valid triangle
2. If valid, classify as:
 - Equilateral
 - Isosceles

- Scalene

Q 6 Time-Based Greeting

Input hour (0–23):

- 5–11 → Good Morning
- 12–16 → Good Afternoon
- 17–20 → Good Evening
- Else → Good Night

Q 7 Password Strength Checker

Input a password and check:

- Length ≥ 8
- Contains digit
- Contains uppercase letter

Print Weak / Medium / Strong

(Use only conditional statements, no loops)

Q 8 Income Tax Calculator

Input annual income:

- $\leq 2.5\text{L}$ → No tax
- 2.5L–5L → 5%
- 5L–10L → 20%
- 10L → 30%

Calculate tax using conditions.

Q 9 Coordinate Position Finder

Input (x, y):

- Origin

- X-axis
 - Y-axis
 - Quadrant I / II / III / IV
-

Q 10 ATM Cash Withdrawal

Input balance & withdrawal amount:

- Amount $\leq$ balance
- Amount is multiple of 100
- Minimum balance after withdrawal $\geq$ 500

Print Approved / Denied.

Q 1 1

Predict output:

`x = 10`

`if x > 5:`

`print("A")`

`if x > 15:`

`print("B")`

`else:`

`if x > 8:`

`print("C")`

`else:`

`print("D")`

Q 1 2

Predict output:

`x = ""`

`if x == False:`

```
    print("Yes")
else:
    print("No")
```

Q 1 3

Predict output:

```
x = None
if x:
    print("True")
elif x == None:
    print("None")
else:
    print("False")
```

Q 1 4

Predict output:

```
x = 5
if x == 5:
    if x > 3:
        if x < 10:
            print("Valid")
```

Q 1 5

Predict output:

```
x = 0
if x == False:
    print("Zero")
if not x:
```



```
print("Not Zero")
```

Q 1 6 Online Order Discount

Apply discount:

- Order $\geq$ ₹5000 $\rightarrow$ 20%
 - Order $\geq$ ₹3000 $\rightarrow$ 10%
 - Order $\geq$ ₹1000 $\rightarrow$ 5%
 - Else $\rightarrow$ No discount
-

Q 1 7 Mobile Data Usage Alert

Input data usage (GB):

- $\leq$ 1GB $\rightarrow$ Normal
 - $\leq$ 2GB $\rightarrow$ Warning
 - 2GB $\rightarrow$ Limit Exceeded
-

Q 1 8 Movie Ticket Pricing

Price based on:

- Age $<$ 12 $\rightarrow$ ₹100
 - Age 12–60 $\rightarrow$ ₹200
 - Age $>$ 60 $\rightarrow$ ₹150
 - Weekend $\rightarrow$ +₹50
-

Q 1 9 Login Attempt System

- Correct ID + password $\rightarrow$ Success
 - Correct ID + wrong password $\rightarrow$ Wrong Password
 - Wrong ID $\rightarrow$ Invalid User
-

Q 2 0 Grading with Bonus Marks

- **Base grade from marks**
- **+5 bonus if attendance $\geq 90\%$**
- **Grade must not exceed A**